1_10xPBMC_Stellarscope_pseudobulk

August 31, 2026

[1]:
```
library(Seurat)
library(ggplot2)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
library(reshape2)
library(clustree)
getwd()
dir.create("figures_10xPBMC_pseudobulk")
dir.create("data_10xPBMC")
dataset_id <- "10xPBMC"
sample <- "pbmc8k"
mode <- "pseudobulk"

colorPBMC <- "#81B29A"

PBMCcelltypeColors <- c("B"="#91bcca",
                "CD14pos_Monocytes"="#3d405b",
                "CD8A"="#f2cc8f",
                "Dendritic"="#c492a7",
                "FCGR3Apos_Monocytes"="#e78063",
                "Megakaryocytes"="#c4b3ab",
                "Memory_CD4_T"="#81b29a",
                "Naive_CD4_T"="#4d7362",
                "NK"= "#88535a")
```

```
Loading required package: SeuratObject

Loading required package: sp


'SeuratObject' was built under R 4.3.3 but the current version is
4.4.1; it is recomended that you reinstall 'SeuratObject' as the ABI
for R may have changed

'SeuratObject' was built with package 'Matrix' 1.6.4 but the current
version is 1.7.5; it is recomended that you reinstall 'SeuratObject' as
the ABI for 'Matrix' may have changed
```

```
Attaching package: ‘SeuratObject’


The following objects are masked from ‘package:base’:

    intersect, t


Warning message:
"package ‘dplyr’ was built under R version 4.4.3"

Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:

    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Warning message:
"package ‘RColorBrewer’ was built under R version 4.4.3"
Warning message:
"package ‘reshape2’ was built under R version 4.4.3"
Loading required package: ggraph


Attaching package: ‘ggraph’


The following object is masked from ‘package:sp’:

    geometry
```

'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

```
Warning message in dir.create("figures_10xPBMC_pseudobulk"):
"'figures_10xPBMC_pseudobulk' already exists"
Warning message in dir.create("data_10xPBMC"):
"'data_10xPBMC' already exists"
```

[2]:
```
path <- paste0("/mnt/TEresults2/snakemake_results/results/stellarscope_out/",␣
↪dataset_id, "/", sample, "/", mode, "/")


TEmatrix <- Seurat::ReadMtx(mtx = paste0(path, sample, "_", mode, "-TE_counts.
↪mtx"),
                            cells = paste0(path, sample, "_", mode,␣
↪"-barcodes.tsv"),
                            features = paste0(path, sample, "_", mode,␣
↪"-features.tsv"),
                            feature.column = 1) # read matrix
TEmatrix <- TEmatrix[setdiff(rownames(TEmatrix),"__no_feature"),]
nCells <- ncol(TEmatrix)
thrMinCells <- round(nCells * 0.05)

# create Seurat object with shallow filtering of TEs expressed in at least 50␣
↪cells and cells expressing at least 50 TEs
objTE_allCB <- Seurat::CreateSeuratObject(TEmatrix, project =␣
↪paste0("PBMC_",mode),
                          min.cells = thrMinCells, min.features = 50)

STAR_path <- paste0("/mnt/TEresults/snakemake_results/results/STARoutdir/",␣
↪dataset_id, "/", sample, "/best_Solo.out/Gene")
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1␣
↪# read barcodes seleced by STARsolo


objTE_stellarscope <- objTE_allCB[,filteredBarcodes] # keep only filtered␣
↪barcodes

objTE_stellarscope
```

```
Warning message:
"Feature names cannot have underscores ('_'), replacing with dashes ('-')"
```

```
An object of class Seurat
6368 features across 1188 samples within 1 assay
Active assay: RNA (6368 features, 0 variable features)
 1 layer present: counts
```

[3]:
```
annotation_stellarscope <- read.table("annotation/annotation_stellarscope_hg38.
↪tsv") # generated with annotation_scripts/create_annotations_hg38.Rmd
head(annotation_stellarscope)
```

A data.frame: 6 × 10

	chr <chr>	start <int>	end <int>	sw_score <int>	strand <chr>	locusID <chr>	family <chr>	subfamily <chr>
1	chr1	67108754	67109046	1892	+	L1P5	L1	L1P5
2	chr1	8388316	8388618	2582	-	AluY	Alu	AluY
3	chr1	25165804	25166380	4085	+	L1MB5	L1	L1MB5
4	chr1	33554186	33554483	2285	-	AluSc	Alu	AluSc
5	chr1	41942895	41943205	2451	-	AluY_dup1	Alu	AluY
6	chr1	50331337	50332274	1587	+	HAL1	L1	HAL1

[4]:
```
# Remove some classes of TEs

classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")

stellarscopeTEs <- Features(objTE_stellarscope)

TEsToKeep <- annotation_stellarscope[! annotation_stellarscope$class %in%
 ↪classesToExclude, ]$stellarscopeID

length(setdiff(stellarscopeTEs, TEsToKeep))/ length(stellarscopeTEs) * 100 #
 ↪percentage removed

objTE_stellarscope <- objTE_stellarscope[intersect(stellarscopeTEs, TEsToKeep),]
```

0.0157035175879397

[5]:
```
feature_metadata <- annotation_stellarscope[match(Features(objTE_stellarscope),
 ↪annotation_stellarscope$stellarscopeID),]

head(feature_metadata)
```

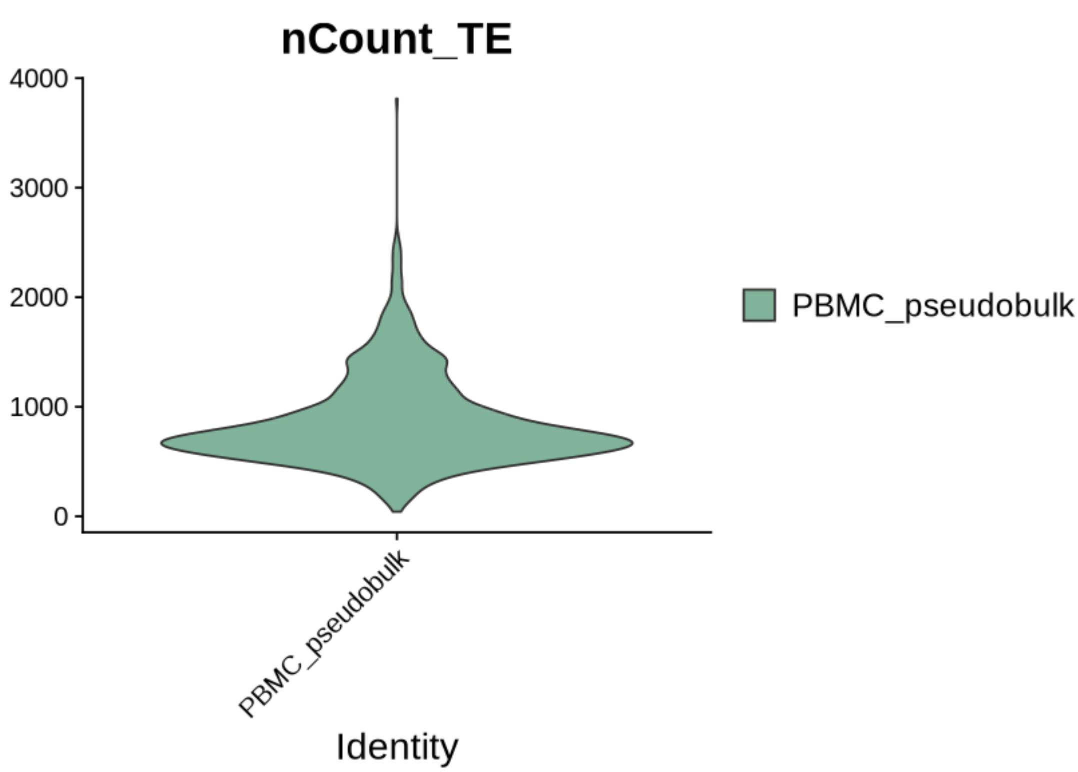
A data.frame: 6 × 10

	chr <chr>	start <int>	end <int>	sw_score <int>	strand <chr>	locusID <chr>	family <chr>	s <
269	chr1	43384732	43384958	1005	-	MIRb_dup9	MIR	M
584	chr1	150732607	150732908	2155	-	AluSz6_dup5	Alu	A
2217	chr1	958244	958549	2303	-	AluSx1_dup51	Alu	A
2515	chr1	1271693	1272004	2458	+	AluSp_dup39	Alu	A
2624	chr1	1394468	1394776	2377	+	AluSx_dup60	Alu	A
2752	chr1	1473369	1473495	1130	-	AluY_dup81	Alu	A

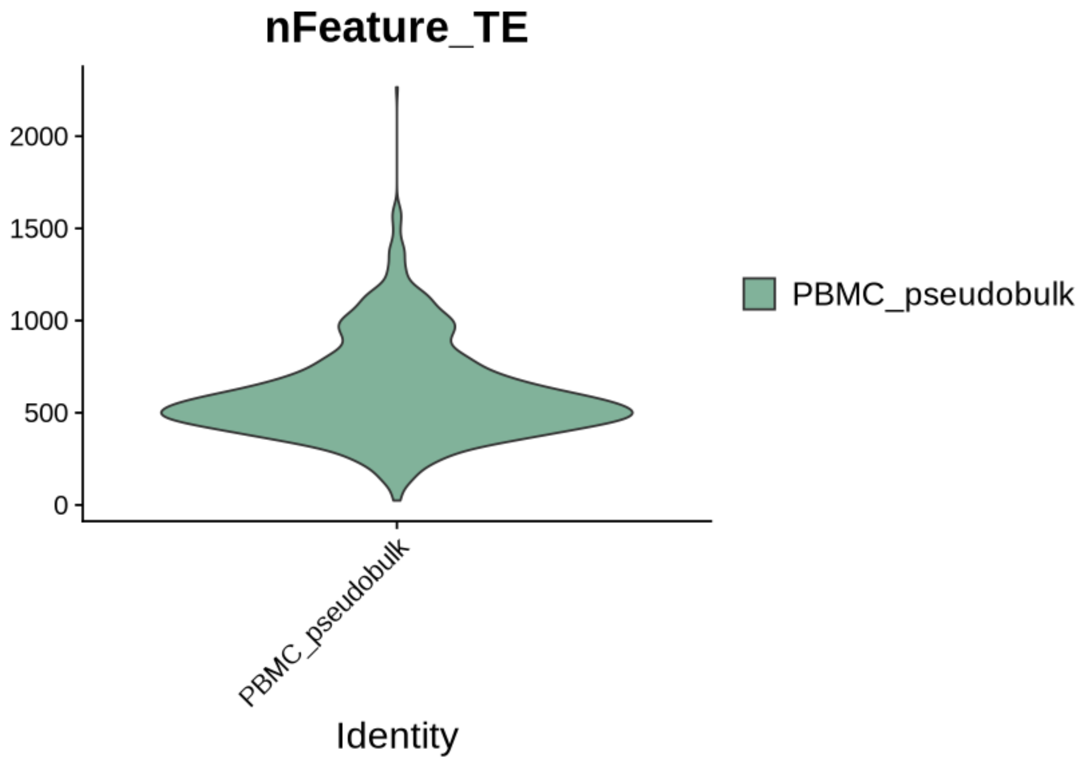
[20]:
```
options(repr.plot.width=7, repr.plot.height=5)


objTE_stellarscope@meta.data$nCount_TE <- objTE_stellarscope@meta.
 ↪data$nCount_RNA
objTE_stellarscope@meta.data$nFeature_TE <- objTE_stellarscope@meta.
 ↪data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_stellarscope, features = c("nCount_TE"), ncol = 1,
```

```
        cols = colorPBMC, pt.size = 0) + theme(text=element_text(size=17))

ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_stellarscope_",mode,".
 ↪png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_stellarscope_",mode,".
 ↪pdf"), device='pdf')

VlnPlot(objTE_stellarscope, features = c("nFeature_TE"), ncol = 1,
        cols = colorPBMC, pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_stellarscope_",mode,".
 ↪png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_stellarscope_",mode,".
 ↪pdf"), device='pdf')
```

```
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Saving 7 x 7 in image
Saving 7 x 7 in image
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."
```



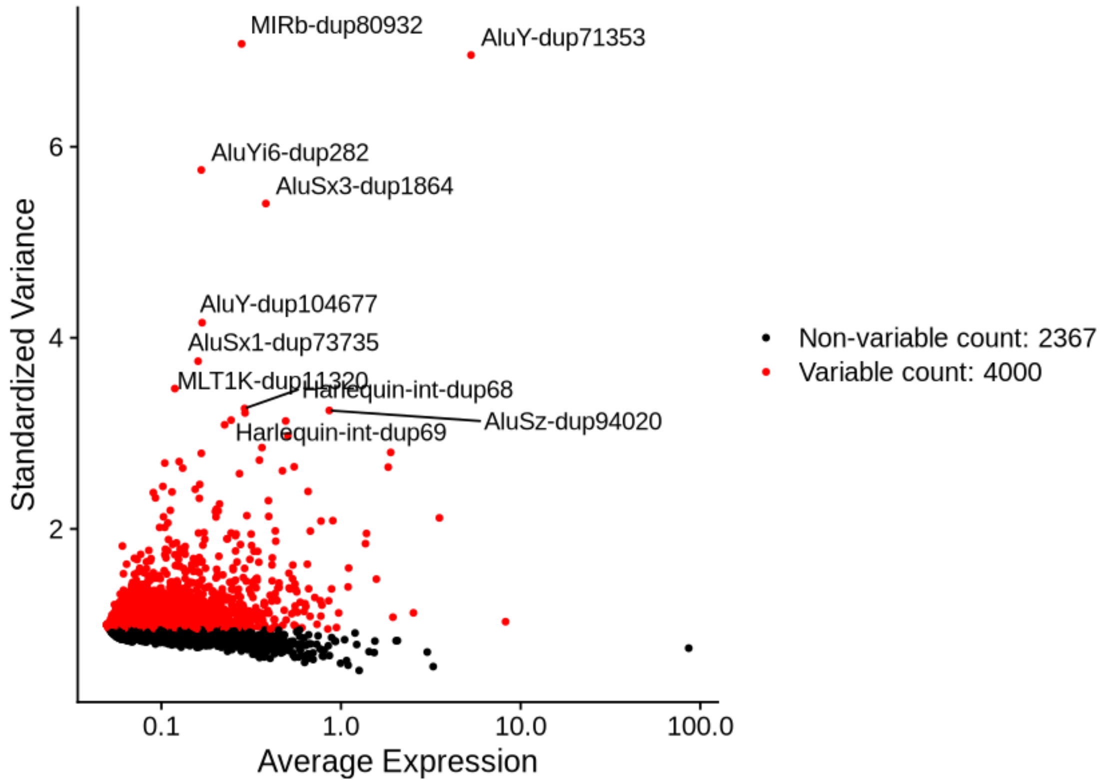
```
Saving 7 x 7 in image
Saving 7 x 7 in image
```



[21]:
```
objTE_stellarscope <- JoinLayers(objTE_stellarscope)

objTE_stellarscope <- NormalizeData(objTE_stellarscope, normalization.method =␣
 ↪"LogNormalize", scale.factor = 10000)

objTE_stellarscope <- FindVariableFeatures(objTE_stellarscope, selection.method␣
 ↪= "vst", nfeatures = 4000)

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_stellarscope), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_stellarscope)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

```
Normalizing layer: counts
```

```
Finding variable features for layer counts

When using repel, set xnudge and ynudge to 0 for optimal results
```



[22]:
```r
gc()
all.genes <- rownames(objTE_stellarscope)
objTE_stellarscope <- ScaleData(objTE_stellarscope) # on hvgs
```

A matrix: 2×6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	18145852	969.1	48286850	2578.8	33894421	1810.2
Vcells	112105818	855.3	295776586	2256.6	295304135	2253.0

```
Centering and scaling data matrix
```

[23]:
```r
objTE_stellarscope <- RunPCA(objTE_stellarscope, features =
 VariableFeatures(object = objTE_stellarscope))

DimPlot(objTE_stellarscope, reduction = "pca") + NoLegend()

ElbowPlot(objTE_stellarscope)
```

PC_ 1
Positive: AluSz6-dup5, MIR3-dup81421, AluSz-dup85172, AluSz-dup94020, AluSc5-dup6172, L1ME3A-dup871, L2c-dup130701, FLAM-A-dup6099, AluY-dup71353, Tigger1-dup5817
AluJo-dup47192, AluJb-dup126028, MER5C1-dup277, MER39-dup143, AluJo-dup69143, AluSx1-dup84605, AluJb-dup3312, MamRTE1-dup5254, AluSq-dup11392, AluJb-dup42076
MIR-dup162327, AluYa5-dup1820, AluSc8-dup11182, L2a-dup163990, HAL1-dup18700, L1ME3Cz-dup8254, AluSz6-dup7653, AluSp-dup29416, AluSz6-dup29860, AluSp-dup32375
Negative: L2b-dup1981, L2b-dup85510, AluSz-dup65970, AluSx-dup129237, AluSx1-dup96779, AluJb-dup73017, Harlequin-int-dup426, MLT-int-dup273, AluSz-dup65021, AluSx-dup34093
MLT1A-dup5928, AluSp-dup3380, AluSz-dup75971, AluSp-dup36543, AluJb-dup29720, MIRc-dup98292, AluSx3-dup21076, AluSq2-dup15156, MLT1A0-dup16914, AluSz-dup76216
L1PA15-dup2196, L1M4-dup14436, AluJo-dup55958, AluSx1-dup56311, L1ME4b-dup14504, MIRb-dup176657, MER92-int-dup473, L1MA9-dup11994, L1MA7-dup6321, AluSx3-dup2001
PC_ 2
Positive: AluSx1-dup105649, AluSz-dup93905, Harlequin-int-dup68, Harlequin-int-dup69, MIRc-dup14097, AluJr-dup26147, AluSz6-dup32784, L1M5-dup2808, AluSq-dup1994, AluSc-dup20728
Harlequin-int-dup70, AluJr-dup5050, AluSc5-dup6699, MIR-dup126706, L1MEi-dup3689, L1PA11-dup2965, FLAM-C-dup21298, AluY-dup104962, AluSc8-dup18117, AluJr-dup84944
AluSx1-dup33189, L1MC5a-dup13849, AluY-dup61306, AluSx1-dup13093, MIRb-dup204693, AluY-dup12183, AluJr-dup84945, L1M7-dup2288, AluYc3-dup171, AluSq-dup1723
Negative: AluJo-dup47192, AluJo-dup69143, AluSz6-dup29860, AluY-dup32753, AluY-dup30565, AluY-dup71353, L2a-dup163990, Tigger1-dup5817, L2a-dup163989, L1ME3A-dup8534
MER5C1-dup277, AluSp-dup32375, AluSx1-dup107728, FLAM-C-dup14640, AluYe5-dup191, AluSq2-dup8938, AluSq2-dup30938, AluSz-dup30155, AluY-dup106670, AluSx-dup129237
AluSz6-dup44790, MER53-dup1521, FLAM-C-dup19077, MER39-dup143, AluSx3-dup20249, AluSg-dup5590, AluSz-dup102545, AluY-dup104677, AluSx1-dup54140, FLAM-A-dup6099
PC_ 3
Positive: L2b-dup85510, L2b-dup1981, AluSx1-dup39707, AluSq-dup8331, AluSp-dup36543, AluSx1-dup88383, MLT-int-dup273, AluSc8-dup16056, AluSx4-dup3994, AluSz-dup76216
MLT1A0-dup16914, AluSz-dup65021, AluSq2-dup15156, AluJr-dup27921, AluSz-dup65970, Charlie4a-dup41, AluSz-dup41379, AluSx-dup100843, AluSx1-dup46526, AluSx-dup54551
AluJb-dup8078, AluSc8-dup19219, AluY-dup44374, AluSx-dup70127, AluJb-dup73017, AluSx1-dup46199, AluJb-dup60468, AluJr-dup4539, MIRb-dup176657, FAM-dup2206

Negative: AluSx1-dup105649, AluSz-dup93905, L1MC5a-dup13849, AluSc5-dup6699, L1M5-dup2808, AluJr-dup84945, AluJr-dup84944, AluY-dup104962, AluSx1-dup86532, ORSL-dup906

AluJr-dup5050, MLT1A0-dup938, AluSz6-dup32784, AluSq-dup1994, MIRc-dup14097, MIRb-dup204693, AluSx-dup6825, MIRc-dup97273, AluSx1-dup103159, MIR-dup126706

AluJb-dup86231, AluSq2-dup14424, AluSx1-dup13093, Tigger9a-dup49, L1MB5-dup1608, Harlequin-int-dup68, L1M5-dup40421, AluY-dup53734, Harlequin-int-dup69, AluSx-dup27178

PC_ 4

Positive: Harlequin-int-dup69, Harlequin-int-dup68, FLAM-C-dup21298, Harlequin-int-dup70, AluJr-dup26694, AluJr-dup72756, MLT1H1-dup1339, MLT1D-dup369, AluY-dup82338, AluSz-dup73882

AluSp-dup49145, MIR-dup154916, AluYe5-dup184, AluSq2-dup58238, AluSx-dup914, AluJb-dup105615, AluSq2-dup49749, L2a-dup77267, AluSz-dup87950, AluSp-dup17244

AluY-dup30565, MIR-dup126706, AluY-dup91356, AluJb-dup75649, AluY-dup43551, AluSx4-dup7418, L1MA3-dup6465, AluSx1-dup40082, MamRep434-dup1164, AluJr-dup33220

Negative: L1MC5a-dup13849, AluSz6-dup22130, AluSc5-dup6699, AluJr-dup84944, AluY-dup104962, L3-dup2815, AluSx1-dup86532, AluJr-dup82212, AluJr-dup84945, AluJb-dup43713

ORSL-dup906, L1M5-dup2808, AluJr-dup5050, MER41A-dup2380, Tigger9a-dup49, MIRc-dup97273, L2a-dup163989, AluSx-dup6825, AluSx3-dup20134, SVA-A-dup189

AluJb-dup86231, AluSc-dup32567, L2a-dup163990, AluJb-dup76256, L1MA9-dup11533, L1M5-dup40421, AluSx1-dup84605, L2c-dup71375, AluSx1-dup84603, AluSg4-dup7103

PC_ 5

Positive: MamRTE1-dup6198, AluJb-dup36574, AluSx1-dup109068, AluJb-dup43713, AluSx1-dup30337, AluSq2-dup15156, AluSp-dup32315, L2b-dup26840, MLT1A0-dup19532, AluY-dup30565

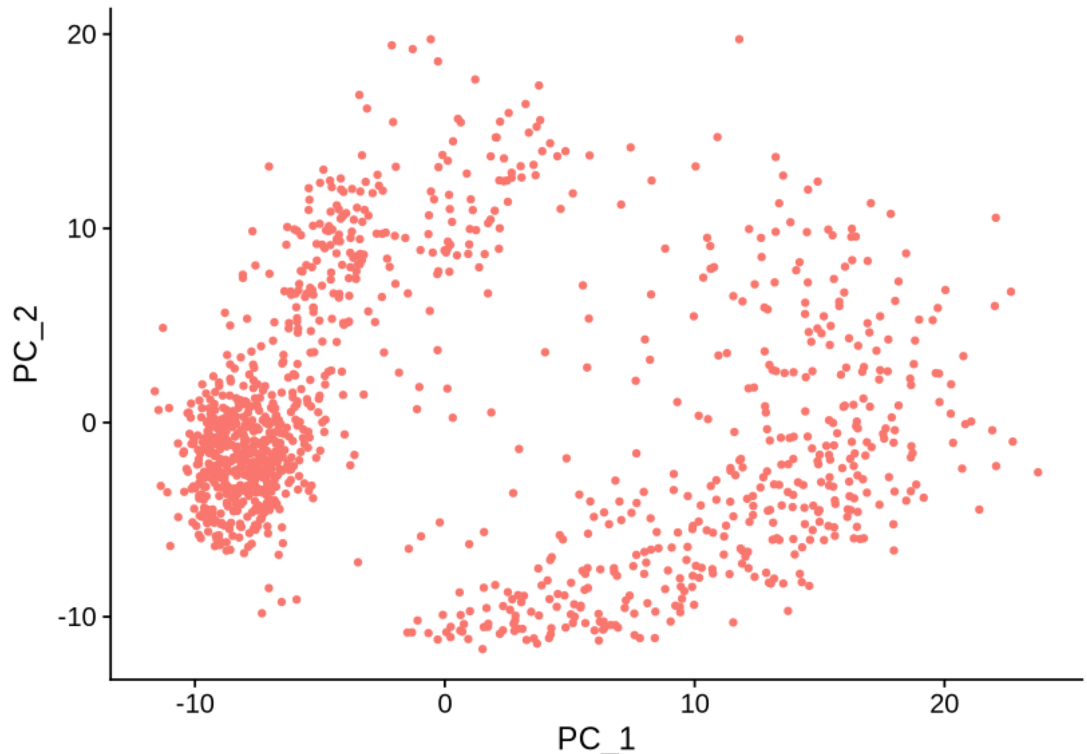
AluSg-dup34300, AluSz-dup88637, AluJo-dup73332, AluJo-dup49220, L1PA15-dup2196, AluY-dup69104, AluSp-dup32314, AluSx1-dup71941, FLAM-A-dup9654, AluSc8-dup14336

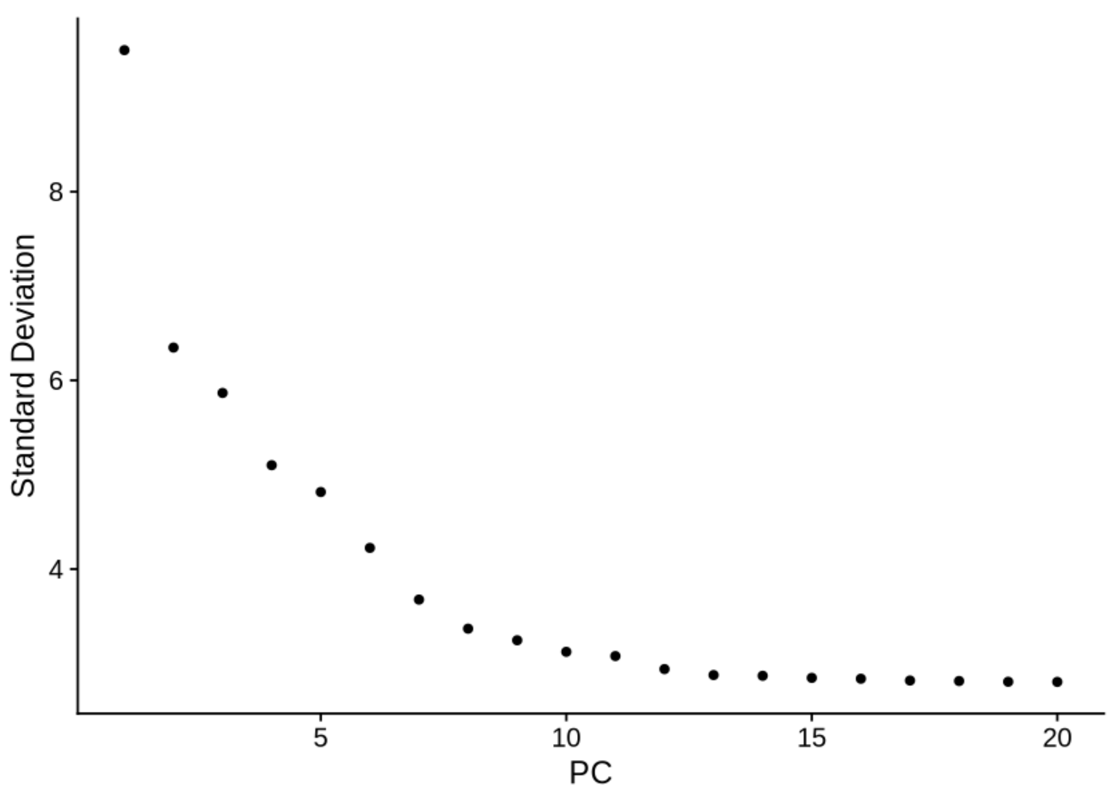
AluSx1-dup56311, MIR-dup120197, AluSq2-dup8706, AluJb-dup79818, AluSg7-dup5445, AluSx1-dup73503, AluY-dup71353, MLT1J-dup10165, AluSp-dup22992, AluSx-dup32527

Negative: L3-dup2815, MIRc-dup98292, AluSx1-dup65945, AluSz-dup84914, MER30-dup2626, Tigger3b-dup268, AluY-dup13774, AluSz-dup65968, AluSx-dup44855, AluSx-dup129237

AluSg-dup30170, AluJb-dup5937, AluSx1-dup73638, AluJo-dup61959, MLT-int-dup273, AluSq2-dup924, L3-dup40489, AluSp-dup12283, AluSq2-dup24157, L2a-dup142356

AluSx1-dup96779, AluSz-dup65970, AluSx1-dup84522, Ricksha-a-dup10, AluY-dup5907, L2c-dup100575, AluY-dup65449, AluSz-dup36471, L1MCa-dup1777, AluSx1-dup88436


PC_2
PC_1
20
10
0
-10
-10
0
10
20


Standard Deviation
PC
8
6
4
5
10
15
20

[1]:
```
objTE_stellarscope <- FindNeighbors(objTE_stellarscope, dims = 1:13, k.param =␣
↪20)
objTE_stellarscope <- FindClusters(objTE_stellarscope, resolution = 1.2)
objTE_stellarscope <- RunUMAP(objTE_stellarscope, dims = 1:13)
DimPlot(objTE_stellarscope, reduction = "umap")
```

```
Error in FindNeighbors(objTE_stellarscope, dims = 1:13, k.param = 20): could no␣
↪find function "FindNeighbors"
Traceback:
```

[25]:
```
FeaturePlot(objTE_stellarscope, reduction = "umap", features = "nCount_TE", pt.
↪size = 0.5) +
  theme_void() +
  theme(text=element_text(size=20))
```

[26]:
```
celltypes <- read.table("/mnt/TEdata/TEbenchmarking/data/whitelists/
↪celltype_annotation_sub_min.tsv")
```

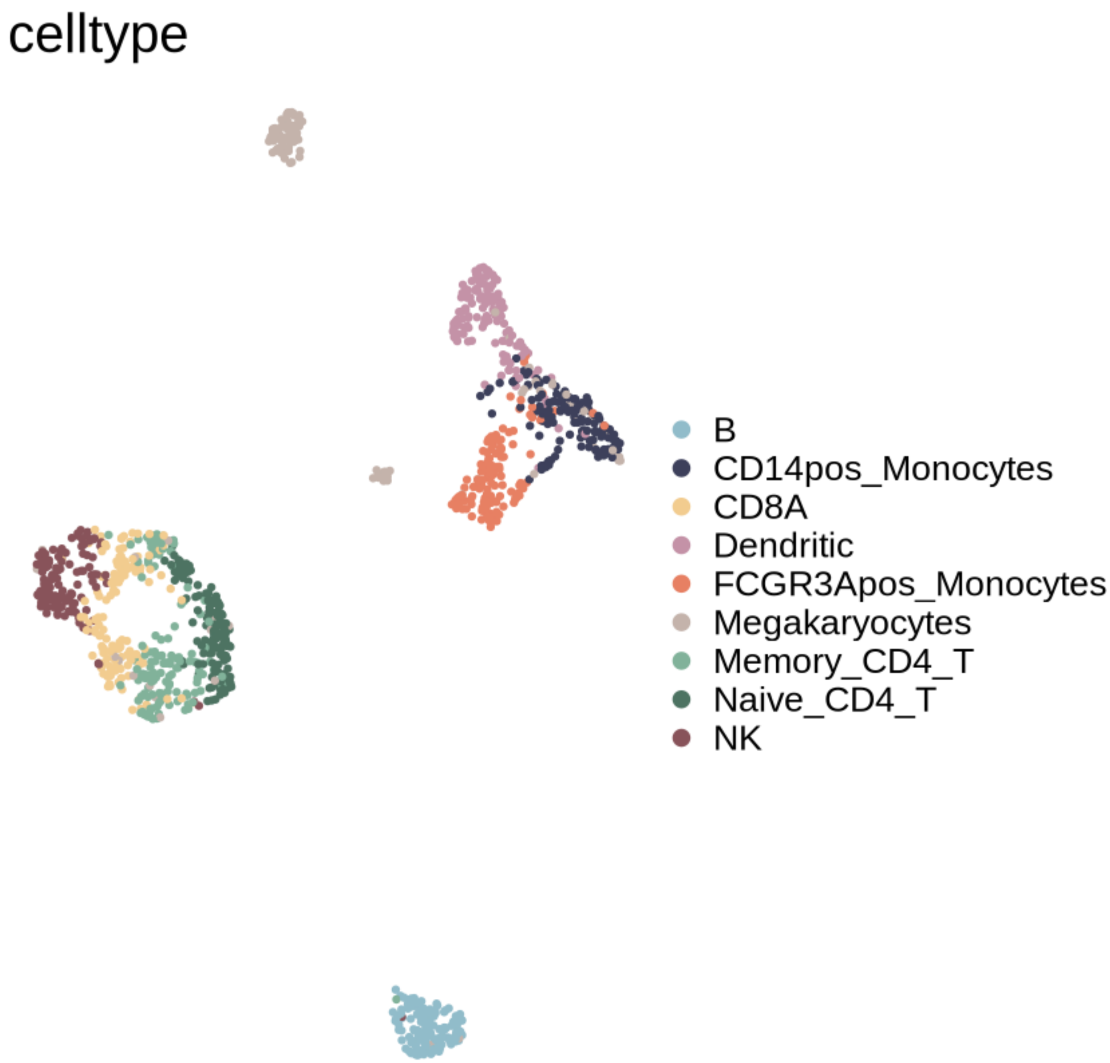
```
table(Cells(objTE_stellarscope) %in% celltypes$V1)

objTE_stellarscope$celltype <- celltypes$V2[match(Cells(objTE_stellarscope),
 ↪celltypes$V1)]
table(objTE_stellarscope$celltype)
```

```
TRUE
1188
```

```
                 B   CD14pos_Monocytes                CD8A          Dendritic
               132                 132                 132                132
FCGR3Apos_Monocytes      Megakaryocytes         Memory_CD4_T        Naive_CD4_T
               132                 132                 132                132
                NK
               132
```

[27]:
```
options(repr.plot.width=7, repr.plot.height=7)

DimPlot(objTE_stellarscope, reduction = "umap", group.by = "celltype",
        cols=PBMCcelltypeColors,
        shuffle=T, pt.size = 1) +
  theme_void() +
  theme(text=element_text(size=20))
ggsave(paste0("figures_",dataset_id,"/
 ↪umap_TEs_locus_celltypes_stellarscope_",mode,".png"), device = "png")
ggsave(paste0("figures_",dataset_id,"/
 ↪umap_TEs_locus_celltypes_stellarscope_",mode,".pdf"), device = "pdf")
```

```
Saving 7 x 7 in image
Saving 7 x 7 in image
```

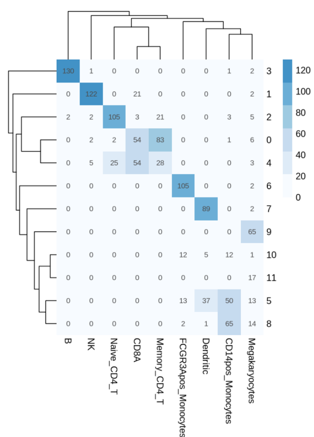


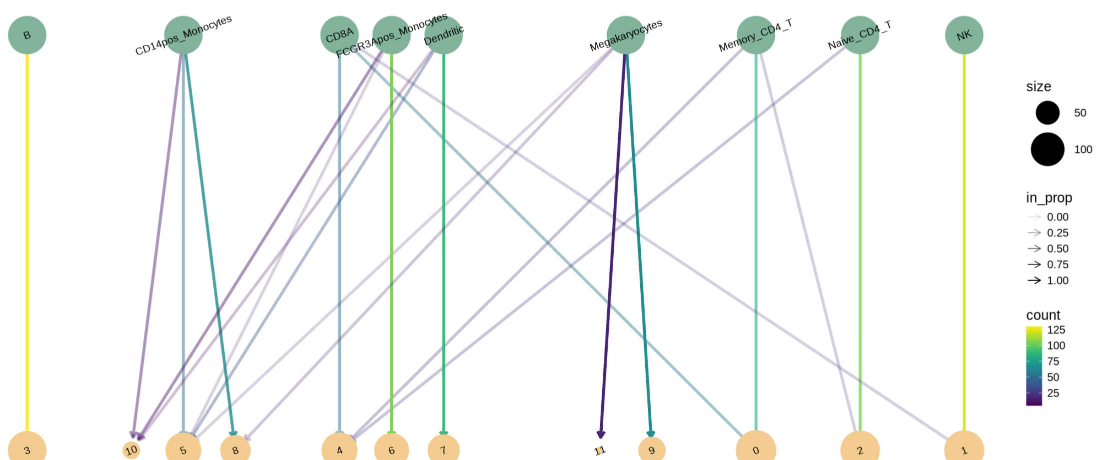
[28]:
```
saveRDS(objTE_stellarscope, paste0("data_", dataset_id, "/stellarscope_",␣
↪dataset_id, "_", mode, "seuratObj.RDS"))
```

[35]:
```
options(repr.plot.width=17, repr.plot.height=7)

# Add cluster information from objTE_clustered to objGenes_clustered
objTE_stellarscope$clusters.0.2 <- objTE_stellarscope$seurat_clusters
objTE_stellarscope$clusters.0.1 <- objTE_stellarscope$celltype

concordanceTable <- table(objTE_stellarscope$seurat_clusters,␣
↪objTE_stellarscope$celltype)
```

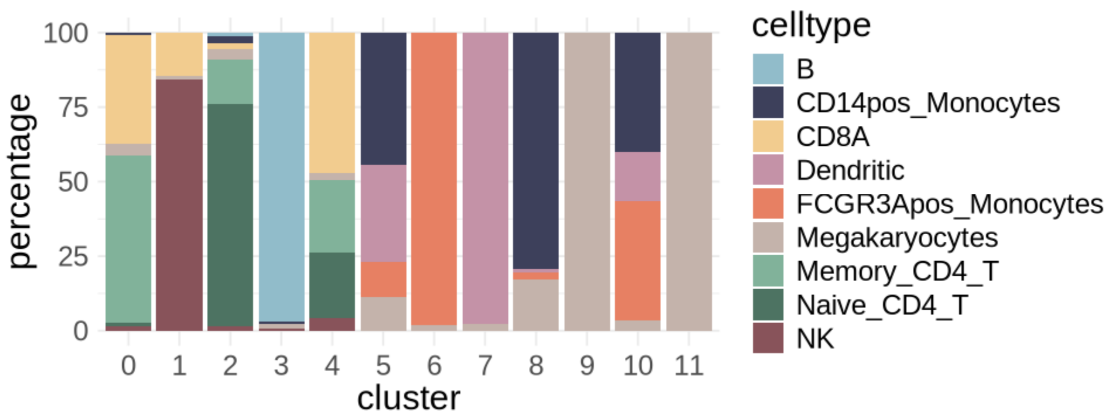
```r
pheatmap(concordanceTable, display_numbers = T, color = brewer.pal(9,'Blues')[1:6],
         border_color = NA, number_format = "%.0f", cellwidth = 25, cellheight = 25)

# Generate the clustree plot
clustree(objTE_stellarscope, prefix = "clusters.", node_text_angle=20, node_text_size=4) + theme(text=element_text(size=15)) +
  scale_color_manual(values=alpha(c("#81B29A","#F2CC8F"), 0.6)) + scale_size(range = c(3,20)) +
  guides(colour = FALSE)
```

```
Scale for size is already present.
Adding another scale for size, which will replace the existing scale.
```





[37]:
```
options(repr.plot.width=8, repr.plot.height=3)

df <- melt(concordanceTable)
df <- df[df$value!=0,]
colnames(df) <- c("cluster","celltype","nCells")
df$cluster <- as.character(df$cluster)
df$clusterSize <- table(objTE_stellarscope$seurat_clusters)[df$cluster]
df$percentage <- as.numeric(df$nCells / df$clusterSize *100)

df$cluster <- factor(df$cluster, levels=c(0,1:length(unique(df$cluster))))


ggplot(df, aes(x=cluster, y=percentage, fill=celltype)) +
  geom_col() +
  scale_fill_manual(values = PBMCcelltypeColors)+
  theme_minimal() + theme(text=element_text(size=18))
ggsave(paste0("figures_10xPBMC/clusterIdentity_barplot_Stellarscope,",mode,".
↪pdf"), width=8, height=3)
```



[ ]: